

# **EXPERIMENTS**

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COURSE CODE:DSA0602 DATA HANDLING AND  
VISUALIZATION

## **SET 1 – PRODUCT INVENTORY MANAGEMENTDATASET**

| Product ID | Product Name | Quantity Available |
|------------|--------------|--------------------|
| 1          | Product A    | 250                |
| 2          | Product B    | 175                |
| 3          | Product C    | 300                |
| 4          | Product D    | 200                |
| 5          | Product E    | 220                |

## **QUESTION 1**

**Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.**

### **AIM**

**To create a bar chart to visualize the quantity available for each product.**

### **PROCEDURE**

- Create the product inventory data.
- Store product names and quantities.
- Use the barplot() function.
- Label the X-axis and Y-axis.
- Give a suitable title to the chart.
- Display the bar chart.

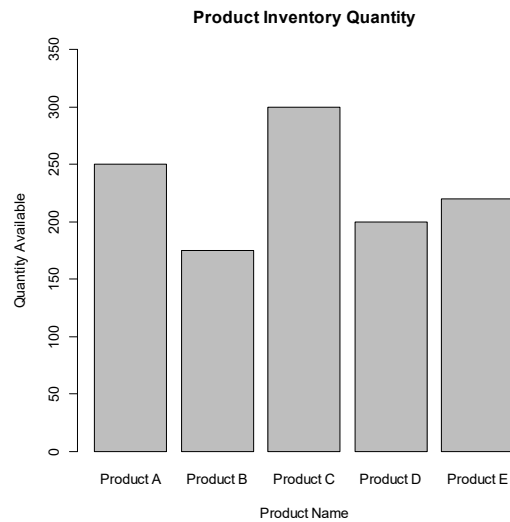
### **CODE**

```
product_name <- c("Product A", "Product B", "Product C", "Product D", "Product E")  
quantity <- c(250, 175, 300, 200, 220)
```

```
barplot(  
  quantity,  
  names.arg = product_name,  
  xlab = "Product Name",
```

```
ylab = "Quantity Available",  
main = "Product Inventory Quantity",  
ylim = c(0, 350)  
)
```

## OUTPUT



## RESULT

Thus, the bar chart was successfully created to visualize the quantity available for each product.

## QUESTION 2

**Generate a stacked bar chart to show the quantity of each product within different product categories.**

### AIM

To create a stacked bar chart showing product quantities according to different product categories.

### PROCEDURE

1. Create product names, categories, and quantities.
2. Create a data frame.
3. Arrange the data according to categories.
4. Use the `xtabs()` function to create category-wise data.
5. Use the `barplot()` function to create the stacked bar chart.
6. Add labels and a legend.

### CODE

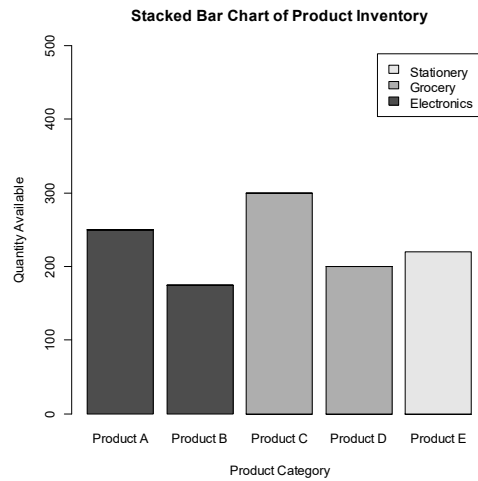
```
product_name <- c("Product A", "Product B", "Product C", "Product D", "Product E")
category <- c("Electronics", "Electronics", "Grocery", "Grocery", "Stationery")
quantity <- c(250, 175, 300, 200, 220)
```

```
inventory <- data.frame(
  Product_Name = product_name,
  Category = category,
  Quantity = quantity
)
```

```
stacked_data <- xtabs(
  Quantity ~ Category + Product_Name,
  data = inventory
)
```

```
barplot(
  stacked_data,
  beside = FALSE,
  xlab = "Product Category",
  ylab = "Quantity Available",
  main = "Stacked Bar Chart of Product Inventory",
  legend.text = TRUE,
  ylim = c(0, 500)
)
```

### OUTPUT



## RESULT

Thus, the stacked bar chart was successfully created to represent product quantities within different categories.

## QUESTION 3

**Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.**

### AIM

To create a scatter plot to study the relationship between product price and quantity available.

### PROCEDURE

1. Create product names.
2. Enter product prices and available quantities.
3. Create a data frame.
4. Use the plot() function.
5. Plot price on the X-axis and quantity on the Y-axis.
6. Add product names to the points.
7. Observe the relationship.

### CODE

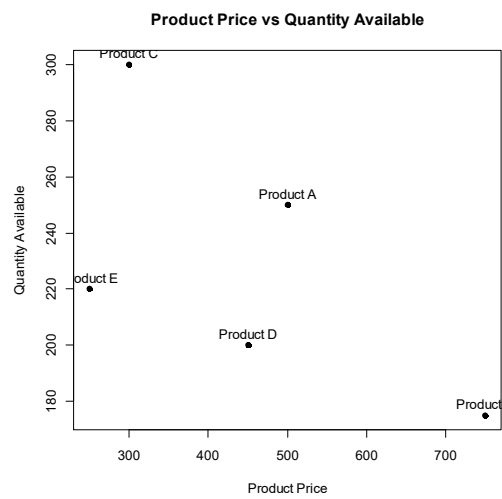
```
product_name <- c("Product A", "Product B", "Product C", "Product D", "Product E")
price <- c(500, 750, 300, 450, 250)
quantity <- c(250, 175, 300, 200, 220)
```

```
inventory <- data.frame(
  Product_Name = product_name,
  Price = price,
  Quantity = quantity
)

plot(
  inventory$Price,
  inventory$Quantity,
  xlab = "Product Price",
  ylab = "Quantity Available",
  main = "Product Price vs Quantity Available",
  pch = 19
)
```

```
text(
  inventory$Price,
  inventory$Quantity,
  labels = inventory$Product_Name,
  pos = 3
)
```

### OUTPUT



## RESULT

Thus, the scatter plot was successfully created to analyze the relationship between product price and quantity available.

## QUESTION 4

**Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.**

### AIM

To develop an interactive Tableau dashboard using the product inventory data.

### PROCEDURE

1. Open **Tableau**.
2. Enter the product inventory data.
3. Create a worksheet for the bar chart.
4. Drag **Product Name** to Columns.
5. Drag **Quantity Available** to Rows.
6. Create another worksheet for the stacked bar chart.
7. Drag **Category** to Columns.
8. Drag **Quantity Available** to Rows.
9. Drag **Product Name** to Color.
10. Create a new dashboard.
11. Add both charts to the dashboard.
12. Add filters for Product Name and Category.
13. Enable interaction between the charts.

### CODE

```
library(shiny)
library(ggplot2)

inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Grocery", "Grocery", "Stationery"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),

  sidebarLayout(
    sidebarPanel(
      selectInput(
        "category",
        "Select Category:",
        choices = c("All", unique(inventory$Category))
      )
    ),
  )
```

```

    mainPanel(
      plotOutput("barChart"),
      plotOutput("stackedChart")
    )
  )
)

server <- function(input, output) {

  filtered_data <- reactive({
    if (input$category == "All") {
      inventory
    } else {
      inventory[inventory$Category == input$category, ]
    }
  })

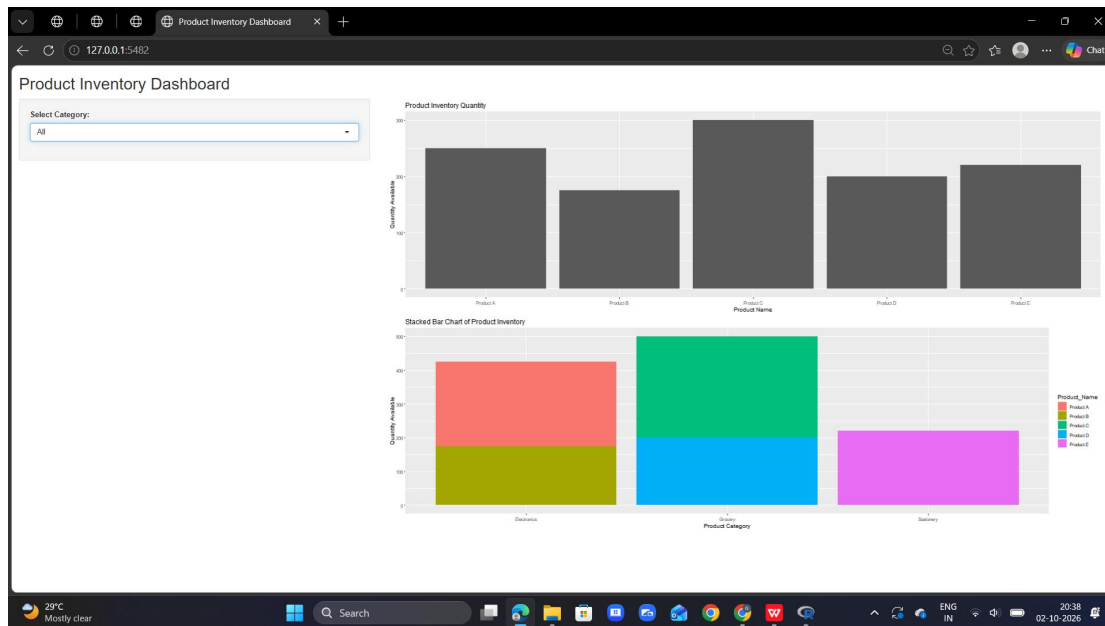
  output$barChart <- renderPlot({
    ggplot(filtered_data(), aes(x = Product_Name, y = Quantity_Available)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Product Inventory Quantity",
        x = "Product Name",
        y = "Quantity Available"
      )
  })

  output$stackedChart <- renderPlot({
    ggplot(filtered_data(), aes(x = Category, y = Quantity_Available, fill =
Product_Name)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Stacked Bar Chart of Product Inventory",
        x = "Product Category",
        y = "Quantity Available"
      )
  })
}

shinyApp(ui = ui, server = server)

```

## OUTPUT



## RESULT

Thus, an **interactive product inventory dashboard** was successfully developed in R using **Shiny**, containing a bar chart and a stacked bar chart.